

SYSTEMS ARCHITECTURE CASE STUDY

Boeing 787 Dreamliner

Innovation, distributed authority, and emergent fragility

Central finding. The 787 program shows that product architecture, organizational architecture, and assurance architecture cannot be designed independently. Technical responsibility may be distributed, but aircraft-level knowledge, integration authority, and safety accountability must remain coherent.

Document	Case study, five discussion questions, and rigorous architectural analysis
Prepared for	Aykut Satici
Date	25 August 2026
Format	US Letter, 1-inch margins

Prepared as a self-contained discussion brief.

Case Study

In the early 2000s, executives at Boeing faced a strategic dilemma. Airbus was gaining ground, fuel prices were rising, and airlines wanted efficiency without increasing aircraft size. The answer became the Boeing 787 Dreamliner – a revolutionary composite aircraft promising 20% better fuel efficiency.

But the innovation was not just technological. It was architectural.

Instead of designing and integrating most systems internally, Boeing restructured the industrial architecture of aircraft manufacturing. Major suppliers across Japan, Italy, and the United States were given unprecedented responsibility – not just to manufacture components, but to design entire subsystems. Wings, fuselage barrels, avionics modules, and battery systems were developed in parallel around the globe.

The vision was elegant: risk-sharing partnerships, reduced development cost, faster integration.

The reality was more complex.

Suppliers struggled with integration standards. Design data arrived incomplete. Manufacturing tolerances differed. Communication delays cascaded. When lithium-ion battery incidents led to aircraft groundings worldwide in 2013, the issue appeared localized – yet investigations revealed deeper systemic coupling problems in certification assumptions, thermal containment, and oversight architecture.

The aircraft eventually succeeded commercially. But the development process transformed aerospace supply chain architecture – exposing how tightly coupled global systems can produce nonlinear failure.

*The Dreamliner did fly.
But the system that built it revealed emergent fragility.*

Discussion Questions

1. Which early architectural decision had the highest systemic leverage: composite materials or globalized subsystem design authority?
2. Where did tight coupling exist, and how did it amplify risk?
3. Was the battery crisis a technical subsystem issue, or an emergent property of distributed governance?
4. How did financial market pressures influence architectural trade-offs?
5. If you were Chief Systems Architect in 2003, what uncertainty mitigation mechanisms would you embed?

Architectural Frame

The 787's central lesson is that product architecture, organizational architecture, and assurance architecture cannot be designed independently. Boeing pursued a highly novel aircraft while simultaneously decentralizing design authority, compressing schedules, and relying on contractual coordination to perform work previously handled through direct engineering control. Each choice was defensible in isolation; their interaction created systemic fragility.

Layer	Major change	Resulting uncertainty
Product	Composite primary structure, more-electric systems, lithium-ion batteries	New failure modes and limited historical evidence
Organization	Suppliers designed and integrated major subsystems	Responsibility dispersed across corporate boundaries
Information	Design evolved concurrently across a global network	Interfaces and configuration data matured late
Governance	Risk-sharing contracts replaced some direct oversight	Commercial accountability did not guarantee technical visibility
Program	Aggressive cost and schedule targets	Problems propagated before they were discovered

1 Which decision had the highest systemic leverage?

Globalized subsystem design authority had greater systemic leverage than the use of composite materials.

Composite construction was a major technical innovation. It introduced difficult questions involving manufacturing consistency, structural inspection, repair, lightning protection, fatigue characterization, and certification. Nevertheless, these risks were concentrated primarily within the product and production architecture. With sufficiently strong internal integration, testing, and process control, they were technically manageable.

Delegating subsystem design authority changed the architecture of the entire development system. It altered:

- who possessed system knowledge;
- who controlled interfaces;
- where design decisions were made;
- how inconsistencies were detected;
- how quickly corrective action could occur;
- whether Boeing could observe lower-tier supplier problems; and

- who was accountable when an issue crossed subsystem boundaries.

This decision therefore affected every technology in the aircraft, including the composite structure. A supplier could optimize its own work package while inadvertently creating consequences for adjacent structures, wiring, thermal management, software, manufacturing, certification, or final assembly.

The key distinction is between *outsourcing production* and *outsourcing architectural authority*. Production can be delegated when designs and interfaces are stable. Delegating design during a highly concurrent, technologically novel program transfers decisions but does not transfer the aircraft integrator's ultimate responsibility for system behavior.

Counterfactual test

- A composite aircraft developed under mature, centrally controlled integration would still have been difficult.
- A conventional aircraft developed through immature, fragmented design governance could still have suffered major integration failures.
- Combining both novelty dimensions made the effects multiplicative rather than additive.

Thus, composites created substantial technical uncertainty; distributed design authority determined whether that uncertainty could be seen and controlled.

2 Where did tight coupling exist, and how did it amplify risk?

Tight coupling existed in several mutually reinforcing forms.

Technical coupling

Aircraft subsystems cannot be completely modular because they share physical and functional resources:

- structural geometry and load paths;
- electrical generation and distribution;
- thermal management;
- software and data buses;
- hydraulics and actuation;
- environmental control;
- weight, center of gravity, and aerodynamic performance; and
- fire containment and emergency behavior.

A change in one subsystem could alter several others. Equipment changes, for example, could affect electrical load, heat rejection, weight, wiring routes, structural attachment, software, and certification evidence.

Interface coupling

Suppliers were expected to design in parallel, but parallel development works only when interfaces are precise, stable, testable, and governed by a clear authority. If interface-control documents are incomplete or changing, each participant makes local assumptions. Those assumptions may remain individually reasonable while being globally incompatible.

Geometric mismatches, incompatible tolerances, incomplete wiring installations, or differing digital definitions are symptoms of an interface architecture that is not mature enough to support full concurrency.

Schedule coupling

The program depended on major sections arriving substantially complete for rapid final assembly. That created low slack between upstream design, fabrication, transport, assembly, testing, and certification.

When an upstream supplier delivered incomplete work, Boeing did not receive merely a late component. It inherited unfinished engineering and manufacturing tasks at the final assembly stage, where correction was slower and more expensive. A delay at one node therefore consumed resources needed to resolve other delays.

Information and configuration coupling

Different suppliers could work from different revisions, assumptions, models, or maturity levels. A technically correct design based on an obsolete interface definition is still wrong at the system level.

Weak configuration synchronization produces nonlinear effects:

1. One interface changes.
2. Several suppliers continue working from older assumptions.
3. Physical hardware is manufactured.
4. Incompatibility appears during integration.
5. Redesign affects already completed work.
6. Rework consumes schedule and engineering capacity.
7. Other reviews are rushed, creating further latent defects.

Governance coupling

Boeing retained responsibility for aircraft-level performance and certification, but much of the detailed knowledge needed to exercise that responsibility resided in supplier organizations. Authority, knowledge, and accountability were therefore not perfectly aligned.

This is a particularly dangerous form of coupling: the integrator is accountable for outcomes it may not be able to observe early enough to control.

Economic coupling

Risk-sharing contracts connected supplier cash flow to program milestones and anticipated production volume. These incentives encouraged participation and reduced Boeing's apparent up-front expenditure, but they could also encourage optimistic reporting, delayed escalation, local cost minimization, or insufficient investment in integration capability.

In combination, these couplings reduced the program's ability to absorb variation. Tight coupling does not automatically produce failure; it becomes dangerous when combined with novelty, concurrency, poor observability, and inadequate reserves.

3 Was the battery crisis technical or emergent?

It was both, but at different causal levels: technical in initiation and systemic in consequence.

At the proximate level, the incidents involved a technical subsystem: lithium-ion cells, electrical behavior, internal failure, thermal runaway, propagation between cells, charging and protection, manufacturing variability, and containment.

It would be inaccurate to claim that distributed governance directly caused an internal short circuit or another electrochemical initiating event. Components can fail even in well-governed programs.

At the system level, however, the crisis exposed an assurance problem. The relevant question was not merely, "Can a battery cell fail?" It was:

Given realistic manufacturing variability and uncertain failure mechanisms, can a single failure propagate into an aircraft-level hazardous condition, and is the evidence for preventing or containing that outcome adequate?

That question crosses several boundaries:

- cell manufacturer;
- battery assembler;
- charging and electrical-system designer;

- aircraft integrator;
- certification authorities;
- safety-analysis and test organizations; and
- operational monitoring and maintenance organizations.

The incidents therefore illustrate the difference between component reliability and system safety. Even if each organization satisfies its assigned requirement, the combined system may remain unsafe if:

- assumptions are inconsistent;
- common-cause failures are underestimated;
- failure independence is assumed rather than demonstrated;
- tests do not represent plausible abuse or manufacturing variation;
- containment is treated as secondary because prevention is considered sufficiently reliable; or
- no single party owns the complete safety argument.

The battery failure was consequently technical in initiation but systemic in consequence. The grounding became an emergent property of the interaction among novel technology, certification assumptions, subsystem integration, failure containment, and distributed oversight.

4 How did financial pressures shape architectural trade-offs?

Financial-market pressure likely operated through several architectural mechanisms rather than through a simple instruction to “cut corners.”

Lower apparent development capital

Risk-sharing suppliers funded portions of development, reducing Boeing’s immediate financial exposure. This made the program appear more capital-efficient, but some cost was converted into coordination risk, reduced visibility, and future recovery expense.

Financial risk was transferred contractually, while system-integration risk remained with Boeing.

Accelerated revenue expectations

A shorter development cycle promised earlier deliveries, customer payments, production learning, and market share. This favored concurrency: design, tooling, supplier development, manufacturing, and certification proceeded simultaneously.

Concurrency can shorten schedules when uncertainty is low. Under high novelty, it creates a large rework multiplier because downstream work begins before upstream decisions stabilize.

Focus on core competencies

The strategy implicitly treated large portions of aircraft development as modular work packages that could be purchased from capable partners. This reduced internal staffing and fixed cost, but it also risked misclassifying systems integration knowledge as non-core.

The architecture of an aircraft may be represented by modules; the knowledge required to integrate those modules is not itself modular.

Optimistic milestone behavior

Aggressive public commitments can distort information flow. Managers and suppliers may hesitate to report uncertain maturity because acknowledging uncertainty threatens schedules, contractual payments, market credibility, or career outcomes.

This creates a serious systems problem: the program's official state becomes more mature than its physical state.

Deferred-cost illusion

Reducing early oversight, integration testing, prototypes, or schedule reserves improves near-term financial metrics. However, late integration defects are vastly more expensive. The organization effectively exchanges visible preventive cost for uncertain downstream liabilities.

The core trade-off was therefore not simply cost versus quality. It was observable present cost versus poorly quantified future system risk.

5 What should a Chief Systems Architect have embedded in 2003?

I would preserve the benefits of global specialization but impose a stronger technical-control architecture. The goal would not be to centralize every design decision; it would be to keep aircraft-level knowledge, interface authority, and assurance responsibility indivisible.

1. Explicit architectural ownership

Boeing should retain direct authority over:

- aircraft-level requirements;
- cross-subsystem trade studies;
- interface definitions;
- safety-critical design assumptions;
- configuration baselines;
- verification architecture;
- certification evidence; and

- acceptance of lower-tier suppliers for critical systems.

Every interface should have one named Boeing technical owner. Shared responsibility without a decision owner is usually unowned responsibility.

2. Maturity-based delegation

Design authority should be delegated progressively, based on demonstrated capability rather than contract scope alone. A supplier would earn increasing autonomy by demonstrating:

- requirements traceability and interface discipline;
- configuration control and modeling consistency;
- manufacturing-process capability;
- verification competence; and
- effective oversight of its own suppliers.

Critical work packages would receive deeper oversight than low-consequence components.

3. Stable interface baselines

The program should establish staged interface baselines:

1. Functional interface baseline.
2. Logical and data interface baseline.
3. Geometric and physical interface baseline.
4. Production interface baseline.
5. Certification and verification baseline.

Changes after each baseline would require impact analysis across all affected parties. Interface readiness, rather than percentage-complete reporting, should control authorization to manufacture.

4. Integrated digital engineering

All partners should operate from a governed system model and authoritative configuration repository, including geometry, requirements, loads, electrical demand, thermal outputs, software and data interfaces, mass properties, verification status, and open assumptions and deviations.

A shared tool is not enough. Data semantics, revision authority, validation rules, and update timing must also be common.

5. Early physical and virtual integration

I would require progressively representative integration environments before flight-test aircraft assembly:

- digital mock-ups;
- hardware-in-the-loop test facilities;
- electrical-power-system rigs;
- thermal and environmental tests;
- structural subcomponent and full-scale tests;
- software integration benches;
- representative battery abuse and propagation tests; and
- production-representative section mating demonstrations.

The principle is to move discovery left: integration failures should occur in controlled test environments, not during final assembly or airline operation.

6. Independent safety-assurance authority

A safety organization should report independently of program cost and schedule leadership. It should own the aircraft-level safety case and challenge assumptions involving:

- failure independence and common-cause failures;
- fault propagation and containment;
- novel materials and energy-storage technologies;
- manufacturing variability and maintenance errors; and
- supplier escape paths.

For novel technology, the burden should be to demonstrate safe failure, not merely low estimated failure probability.

7. An uncertainty register, not merely a risk register

Traditional risk registers often record known events with estimated probability and impact. Novel programs also contain epistemic uncertainty – things the organization does not yet understand well enough to assign meaningful probabilities.

Each critical uncertainty should have:

- an accountable owner;
- evidence needed to retire it;
- a deadline linked to a decision gate;
- a test or analysis plan;
- a fallback architecture; and
- a quantified reserve if it remains unresolved.

Examples would include composite production yield, lightning protection, battery propagation, supplier design maturity, software integration, and global logistics.

8. Architectural margins

The program should carry explicit margins for weight, electrical capacity, thermal capacity, computing resources, structural performance, schedule, engineering labor, test assets, and supplier recovery.

Margins should be centrally controlled and released only through formal trade studies. Using margin to protect an individual subsystem at the expense of system resilience should require aircraft-level approval.

9. Supplier observability down the tiers

Boeing should have direct visibility into safety-critical and schedule-critical lower-tier suppliers, even when they are contractually managed by a Tier 1 partner. This includes:

- design maturity and quality escapes;
- staffing and tool compatibility;
- manufacturing yield and financial health;
- critical single-source dependencies; and
- corrective-action closure.

A Tier 1 supplier cannot be treated as a black box when its internal network contains aircraft-level failure paths.

10. Incentives based on integrated outcomes

Contracts should reward interface maturity, complete engineering data, verified configuration, early disclosure of problems, successful integration tests, production quality, and closed certification evidence.

They should not reward shipment of nominally complete assemblies that transfer unfinished work downstream. Early escalation should be economically safer than concealment or optimism.

11. Realistic program reserves

Schedule and funding reserves should be tied to novelty and coupling. A program introducing composite primary structures, extensive electrical architecture changes, new supply-chain governance, new production methods, and new energy-storage technology should not be planned as though these uncertainties were independent.

Reserves should reflect correlated risk and rework loops, not just the sum of individual risk estimates.

12. Preplanned fallback options

For each high-uncertainty technology or supplier arrangement, the architecture should preserve an alternative:

- alternate materials or manufacturing processes;
- secondary sources for critical components;
- conservative containment designs;
- reduced initial performance targets;
- internal engineering recovery teams; and
- the ability to bring critical design work back under direct control.

Options have value only if preserved before the program needs them.

Overall Assessment

The 787 should not be interpreted as evidence that global outsourcing or technological ambition is inherently unsound. The deeper failure was architectural mismatch.

Boeing adopted:

- high product novelty;
- high organizational distribution;
- high development concurrency;
- high financial pressure; and
- low schedule slack.

Such a system requires exceptionally strong interface governance, technical visibility, independent assurance, and integration testing. Instead, parts of the governance model assumed that contractual partitioning would produce technical modularity.

It did not. The aircraft remained one tightly integrated system even when its development organization was divided among many firms.

Responsibility may be distributed, but aircraft-level knowledge, integration authority, and safety accountability must remain coherent.

End of case study analysis